Arrival-delay prediction with deep learning
Micro-Project 4

https://github.com/ChalesManyang/ANA500-Micro-Projects

Charles Ayuk

National University · ANA500

November 7, 2026

Problem Statement

How accurately can a deep neural network predict arrival delay after a flight departs?

Outcome: Arrival Delay in Minutes, a continuous numerical variable.

A useful estimate could support passenger updates. This project evaluates a retrospective prototype using the supplied survey records.

Hypothesis Formulation

Hypothesis: a neural network can improve arrival-delay RMSE compared with a training-mean baseline and linear regression.

Departure delay should contain most of the predictive information. The network may capture nonlinear relationships with flight distance and passenger travel context.

A small improvement alone may not justify a more complex model.

Acquire

Source: user-supplied airline.csv, read with pandas.read_csv.

129,880 passenger records and 25 original columns. The file contains demographics, travel class, service ratings, delays and a satisfaction category.

No collection dates, airline names or sampling documentation accompany the file. Row order does not establish a time sequence.

Prepare

Exclude 393 rows with unknown arrival outcomes, leaving 129,487 records.

Random seed 42: 90,640 train, 19,423 validation and 19,424 test records. Fit scaling and categorical encoding on training rows only.

Predictors: departure delay, distance, age, gender, customer type, travel type and class. Exclude ID, satisfaction and post-flight service ratings.

No sequence or timestamps exist. Use a feedforward network rather than inventing a recurrent sequence.

Analyze data

Two deep networks: 32–16–8 and 64–32–16 hidden units, ReLU activation and a linear regression output.

Adam optimizer, learning rate 0.001, batch size 256, L2 alpha 0.01. Standardize predictors and the outcome using training statistics.

Train up to 80 epochs. Stop after 8 epochs without validation-RMSE improvement. Restore the best weights and clip predictions at zero.

Validation selects 32–16–8 at epoch 6 (RMSE 9.76 minutes). Compare the frozen model with three baselines on the reserved test set.

Report

Test set: 19,424 records. Error units are minutes.

The network reduces RMSE by 0.10 minute (0.96%) versus linear regression. The departure-delay baseline has the lowest MAE. The evidence gives limited support for added model complexity.

Model	**MAE**	**RMSE**	**R^2**
Training mean	20.76	39.16	-0.000
Departure equals arrival	5.04	10.08	0.934
Linear regression	5.35	10.03	0.934
Deep neural network	5.60	9.93	0.936

Act

Use departure delay as a simple initial estimate. Consider the deep network only if a pilot shows that its small RMSE gain matters operationally.

Collect dated flight-level records with route, weather and flight identifiers. Validate on later flights and monitor large-delay errors before customer-facing use.

The random passenger split cannot demonstrate future-flight performance. Shared-flight rows may inflate apparent generalization.

The findings concern prediction. They do not identify causes of delay.